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Published: 11 October 2024

Citation in BibTeX format

ICACS 2024: 2024 the 8th International Conference on Algorithms, Computing and Systems
October 11 - 13, 2024
NB, Hong Kong, China

RESEARCH-ARTICLE

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ORESTIS PAPADIMITRIOU, University of the Aegean, Mytilene, North Aegean, Greece

ATHANASIOS KANAVOS, University of the Aegean, Mytilene, North Aegean, Greece

MANOLIS MARAGOUDAKIS, Ionian University, Kerkira, Ionian Islands, Greece

VASSILIS C GEROGIANNIS, University of Thessaly, Volos, Thessaly, Greece

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ICACS '24: Proceedings of the 2024 8th International Conference on Algorithms, Computing and Systems (October 2024)
https://doi.org/10.1145/3708597.3708605
ISBN: 9798400718304

A Comparative Study of Sentiment Analysis on Twitter and Reddit Using Deep Learning Techniques

Orestis Papadimitriou
University of the Aegean
Samos, Greece
icsdd20016@icsd.aegean.gr

Athanasios Kanavos
University of the Aegean
Samos, Greece
icsdd20017@icsd.aegean.gr

Manolis Maragoudakis
Ionian University
Corfu, Greece
mmarag@ionio.gr

Vassilis C. Gerogiannis
University of Thessaly
Volos, Greece
vgerogian@uth.gr

Abstract

Twitter and Reddit, as prominent social media platforms, offer rich datasets for sentiment analysis, a key area of text mining that interprets users' opinions and emotions from textual data. This paper addresses the scalability challenges commonly associated with traditional machine learning approaches in sentiment analysis by employing advanced deep learning techniques. Specifically, we explore the efficacy of Convolutional Neural Networks (CNNs), adjusting architectural parameters like the number of convolutional layers, filter sizes, and the number of filters to enhance model performance. Our study improves the accuracy of sentiment analysis on both Twitter and Reddit. Comparative results demonstrate significant advancements in accuracy and processing efficiency, highlighting the potential of CNNs to revolutionize sentiment analysis in diverse social media contexts.

Keywords

Sentiment Analysis, Natural Language Processing (NLP), Social Media Analytics, Convolutional Neural Networks (CNN), Twitter, Reddit, Text Mining

ACM Reference Format:
Orestis Papadimitriou, Athanasios Kanavos, Manolis Maragoudakis, and Vassilis C. Gerogiannis. 2024. A Comparative Study of Sentiment Analysis on Twitter and Reddit Using Deep Learning Techniques. In *2024 the 8th International Conference on Algorithms, Computing and Systems (ICACS) (ICACS 2024), October 11–13, 2024, Hong Kong, China.* ACM, New York, NY, USA, 6 pages. https://doi.org/10.1145/3708597.3708605

1 Introduction

In the digital era, ensuring societal stability necessitates a safe and secure online environment. The proliferation of social media platforms has led to a surge in the dissemination of hate speech, fostering a hostile digital atmosphere that may precipitate conflicts and societal unrest [31]. Although freedom of speech is a fundamental right, there is an urgent need to mitigate the adverse effects of online hate speech to maintain social harmony. This paper explores advanced computational techniques and sophisticated machine learning models aimed at effectively detecting, analyzing, and mitigating the impact of hate speech on platforms such as Twitter and Reddit [8], thus contributing to a healthier digital discourse.

Opinion mining and sentiment analysis have become invaluable tools for businesses and social scientists alike. Leading technology companies, including Google and Microsoft, have invested heavily in developing sentiment analysis systems to bolster their operations and enhance user experiences [15, 17, 19]. Twitter Sentiment Analysis (TSA) addresses the unique challenges posed by the platform's 140-character limit and the colloquial nature of the language used, which complicates the extraction of meaningful sentiment from tweets [9, 10].

Sentiment mining utilizes Natural Language Processing (NLP) and Information Extraction (IE) to process and interpret vast datasets, extracting sentiments from diverse textual sources. This technique integrates computational linguistics and information retrieval (IR) strategies to determine the sentiment polarity—positive, negative, or neutral—of texts [5, 11, 26, 29]. The proliferation of user-generated content on Twitter and Reddit, ranging from product reviews to political discourse, illustrates the growing influence of online opinions [21]. This volume of data presents significant challenges in content navigation and decision-making for users, necessitating more sophisticated analytical tools to manage this information effectively.

Our research specifically aims to improve sentiment analysis techniques for Twitter and Reddit by developing models that can more accurately assess the emotional states and attitudes of users. By focusing on the intense emotional responses that users exhibit towards posts, comments, or discussions, we seek to uncover underlying patterns that predict user engagement and sentiment. This paper presents novel deep learning approaches, particularly enhanced Convolutional Neural Network (CNN) architectures, to achieve higher accuracy in sentiment classification. We compare these models against existing machine learning algorithms to highlight their efficacy and potential for broader application in real-world scenarios.

Moreover, the dynamic nature of language on social media, characterized by evolving slang, emojis, and abbreviations, poses additional challenges for traditional sentiment analysis models that

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ICACS 2024, Hong Kong, China
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ACM ISBN 979-8-4007-1830-4/24/10
https://doi.org/10.1145/3708597.3708605

often rely on static lexicons or simplistic algorithms. Our approach incorporates state-of-the-art deep learning techniques that adapt more effectively to these linguistic variations. By implementing and refining CNN architectures specifically tuned for the nuances of social media text, our research not only addresses the challenge of real-time sentiment tracking but also enhances the ability to discern subtle contextual cues that are often overlooked by conventional models. Our approach also includes the development of a dynamic training framework that continuously updates the model's parameters to reflect the latest linguistic trends, ensuring that our system remains effective in the ever-changing landscape of social media communications. Furthermore, this paper details the application of transfer learning techniques, which leverage pre-trained models on large datasets to improve performance on domain-specific tasks, thereby reducing the need for extensive labeled data in sentiment analysis.

The anticipated outcome of this research is not only technological advancement but also a social one. By improving the accuracy and responsiveness of sentiment analysis, we can better identify and mitigate harmful content, potentially reducing the spread of hate speech and its corrosive effects on society. Enhanced sentiment analysis tools will empower platform moderators and policymakers with more effective strategies for fostering a positive online culture, ultimately contributing to a more informed and respectful digital discourse. The potential of these advanced methodologies to significantly elevate the accuracy and responsiveness of sentiment analysis systems offers profound implications for the future of digital communications management, particularly in areas sensitive to public sentiment and social harmony.

The remainder of this paper is organized as follows. Section 2 reviews the related work in the field, highlighting prior studies and advancements in sentiment analysis, particularly those utilizing deep learning methodologies. Section 3 describes our proposed methodology, detailing the specific CNN architectures and configurations employed for sentiment analysis on Twitter and Reddit. Section 4 presents the evaluation of our experiments, analyzing the performance of our model across various metrics and configurations. Finally, Section 5 summarizes the key findings and proposes avenues for further research. Through this structure, the paper aims to provide a comprehensive examination of the enhancements deep learning offers to sentiment analysis in social media contexts.

2 Related Work

The field of sentiment analysis has rapidly evolved with the expansion of digital communication channels, becoming a crucial facet of understanding human behavior in the digital space. This section reviews existing methodologies, advances in sentiment analysis, and its application in diverse fields, highlighting the importance of this research area in processing and interpreting the vast amounts of data generated daily on social media [6].

Sentiment analysis aims to assess the mood or emotional state of individuals by analyzing their reactions to texts, communications, or events. These reactions can be emotionally charged judgments or opinions, reflecting the author's or speaker's feelings, or anticipated emotional responses, intended to influence the audience [18]. As the internet continues to burgeon with user-generated content on a myriad of topics, from product evaluations to entertainment reviews, the challenge of filtering through this data to make informed decisions becomes increasingly significant [23]. Leveraging sentiment analysis to mine and systematically organize this information is essential, offering clearer insights for businesses and consumers alike [2]. The methodologies for detecting and classifying emotions in text are primarily categorized into three approaches: lexicon-based, machine learning-based, and hybrid methods. Lexicon-based methods utilize the intrinsic positive or negative meanings of words, while machine learning approaches view text as data for classification problems, tackled through unsupervised, semi-supervised, or supervised learning [4].

Effective sentiment analysis begins with meticulous data preprocessing, especially when dealing with voluminous text data from platforms like Twitter and Reddit. This process involves normalizing text by converting upper-case letters to lower-case, removing redundant words or links, expanding contractions, eliminating non-alphabet characters or symbols, discarding stop words, and deleting duplicate entries. Further preprocessing steps include tokenization, stemming, lemmatization, and Part of Speech (POS) tagging, which help in refining the data for more precise sentiment analysis [7].

Sentiment analysis's primary goal is to categorize data into positive, neutral, or negative sentiments. However, it also delves into affective computing, encompassing a broad array of emotions. Depending on the context, sentiments in political debates, for example, might be further divided into satisfied or angry [3]. More detailed analyses can classify sentiments into specific emotions like anxiety, sadness, anger, excitement, or happiness, allowing for a deeper understanding of user interactions and reactions [27, 28].

Expanding beyond textual analysis, sentiment analysis applies to audio and visual data through multimodal sentiment analysis. This approach studies expressions, body movements, and vocal tones to generate a more comprehensive analysis of sentiments, broadening the scope of NLP applications beyond traditional text-based data [25, 30]. Advances in neural networks and emerging areas like neurosymbolic AI, which integrates deep learning with symbolic reasoning, further enhance the capabilities of NLP, promising significant strides in our understanding of complex human sentiments [16, 22, 24].

In summary, sentiment analysis not only provides essential insights into individual and collective emotional states but also facilitates the navigation of the complex landscape of online communications. As this field continues to advance, incorporating more sophisticated algorithms and broader data sources, it will play a pivotal role in shaping future technologies and strategies for digital interaction. This research underpins the development of tools that can effectively manage and utilize large-scale data, ensuring more nuanced and actionable insights across various sectors.

3 Proposed Methodology

This section delineates the proposed methodology for enhancing sentiment analysis through the integration of advanced machine learning techniques and Natural Language Processing (NLP). We introduce a sophisticated architecture employing Convolutional

Neural Networks (CNNs) tailored to understand and process language data, distinguishing our approach from conventional applications primarily focused on visual data analysis [12–14, 20]. The proposed methodology involves several innovative components aimed at improving the accuracy and efficiency of sentiment detection in large-scale text data, such as those generated on social media platforms like Twitter and Reddit.

3.1 Natural Language Processing

Natural Language Processing (NLP) is a pivotal subfield of artificial intelligence that focuses on enabling machines to understand, interpret, and generate human language in a way that is both meaningful and efficient. This involves a complex interplay of algorithms and models drawn from linguistics, computer science, and machine learning, particularly deep learning. Key challenges in NLP include syntactic parsing to discern grammatical structures, semantic analysis to understand meanings, and pragmatic processing to interpret the intended messages within context.

Mathematically, NLP models often employ vector space models to transform text into numerical forms, where each dimension represents a feature of the text, such as word frequency or context-based embeddings. For example, the term frequency-inverse document frequency (TF-IDF) and Word2Vec models are common for converting text data into a format suitable for machine learning algorithms.

3.2 Convolutional Neural Networks

Although traditionally associated with image processing, Convolutional Neural Networks (CNNs) have been effectively adapted for text analysis by treating textual input as a one-dimensional spatial domain. In this context, a CNN learns hierarchical representations of textual data by applying multiple convolutional filters to capture local features such as n-grams (sequences of n words).

The mathematical framework for a CNN involves convolution operations that apply filters to the input data, defined as:

$$f(x) = \mathbf{w} \cdot x + b \tag{1}$$

where $\mathbf{w}$ is a weight matrix, x is the input feature map, and b is a bias term. These operations are followed by non-linear activation functions such as ReLU to introduce non-linearities into the model, and pooling layers to reduce dimensionality and enhance feature detection.

3.3 Proposed CNN Architecture for Text Analysis

Our proposed CNN architecture for text analysis comprises an embedding layer followed by several convolutional and pooling layers designed specifically for sequence data. The embedding layer maps each word in the input sequence to a high-dimensional vector space, capturing semantic relationships between words. The network configuration is as follows:

- Input Layer: Handles sequences of 1211 tokens.
- Embedding Layer: Transforms tokens into 20-dimensional vectors, vocab size approx. 49,957.
- Conv1D Layer: 32 filters of size 3, activation = 'relu'.
- MaxPooling1D Layer: Pool size of 3.
- Conv1D Layer: 64 filters, activation = 'relu'.
- MaxPooling1D Layer: Pool size of 3.
- Conv1D Layer: 128 filters, activation = 'relu'.
- GlobalMaxPooling1D Layer: Condenses the feature map across the sequence.
- Output Layer: Dense layer with 3 units for classification (positive, neutral, negative sentiments).

The total number of trainable parameters in this model is 1,032,391. The deep structure of this CNN allows for the extraction and learning of complex patterns in text, significantly enhancing the model's ability to discern nuanced emotional expressions.

Table 1 provides a summary of the CNN architecture, highlighting the output shapes and parameter counts at each layer, which are critical for understanding the complexity and capacity of the model.

Table 1: CNN Architecture for Sentiment Analysis

Layer Type	Output Shape	Parameter Count
Input Layer	(None, 1211)	0
Embedding Layer	(None, 1211, 20)	999,140
Conv1D Layer	(None, 1209, 32)	1,952
MaxPooling1D Layer	(None, 403, 32)	0
Conv1D Layer	(None, 401, 64)	6,208
MaxPooling1D Layer	(None, 133, 64)	0
Conv1D Layer	(None, 131, 128)	24,704
GlobalMaxPooling1D Layer	(None, 128)	0
Dense Layer	(None, 3)	387

3.4 Rationale Behind CNN Architecture

The Convolutional Neural Network (CNN) architecture selected for this study is particularly tailored to address the unique challenges presented by Twitter and Reddit data. These platforms are characterized by their brevity and informal linguistic styles, including the use of slang, emojis, and abbreviations, which require a robust model capable of capturing contextual nuances from limited textual input. Our CNN model employs multiple convolutional and pooling layers, which are ideal for this task as they excel in extracting hierarchical features from spatial data — in our case, the spatial arrangement of words in tweets and comments.

Firstly, the initial convolutional layers capture basic textual features such as n-grams which are crucial for understanding sentiment in short phrases commonly found on Twitter and Reddit. Subsequent layers aggregate these features into more complex representations, allowing the model to discern intricate patterns of sentiment and context that are not immediately apparent. The depth of our model, indicated by the number of convolutional layers, allows for an extensive learning capacity, necessary to handle the wide variety of topics and expressions found in social media discourse.

Moreover, the specific parameters of our CNN architecture, such as filter sizes and the number of filters, were chosen based on preliminary experiments which indicated that smaller filters are

effective at capturing the nuances of short textual data while a higher number of filters increases the model's ability to process diverse linguistic expressions found in social media. This configuration ensures that our model is not only sensitive to the lexical and syntactic variances within the brief text but is also scalable to the voluminous data typical of these platforms.

3.5 Implementation and Training Strategy

The implementation of our CNN model will utilize the TensorFlow and Keras libraries, leveraging GPU acceleration for efficient training. We plan to train the model using a dataset comprising labeled sentiment data from Twitter and Reddit posts. The training process will employ the Adam optimizer for effective error minimization and backpropagation. We will utilize cross-entropy loss as our loss function, suitable for multi-class classification problems.

$$L = -\sum_{i=1}^{C} y_i \log(p_i) \tag{2}$$

where L is the loss, C is the number of classes, y_i is the binary indicator (0 or 1) if class label i is the correct classification, and p_i is the predicted probability of the class.

3.6 Evaluation Metrics

The efficacy of our proposed CNN model will be assessed using three primary metrics: accuracy, loss, and computational efficiency. These metrics are crucial for evaluating the performance and practical applicability of the model in real-world scenarios, where both accuracy and processing speed are critical.

- **Accuracy:** This metric measures the proportion of correct predictions made by the model out of all predictions made, providing a straightforward indicator of model performance.

$$\text{Accuracy} = \frac{\text{Number of Correct Predictions}}{\text{Total Number of Predictions Made}} \tag{3}$$

- **Loss:** We utilize categorical cross-entropy as our loss function, which is particularly effective for classification problems involving multiple classes. The loss provides a measure of how well the model's predicted probabilities match the actual class labels.

$$L = -\sum_{c=1}^{C} y_{o,c} \log(p_{o,c}) \tag{4}$$

where L is the loss for a single observation o, C is the number of classes, $y_{o,c}$ is a binary indicator of whether class label c is the correct classification for observation o, and $p_{o,c}$ is the predicted probability of class c for observation o.

- **Computational Efficiency:** This metric includes the training time per epoch and inference time per sample. It is essential for deploying models in environments where response time and resource utilization are critical.

$$\text{Training Time} = \frac{\text{Total Training Duration}}{\text{Number of Epochs}} \tag{5}$$

$$\text{Inference Time} = \frac{\text{Total Time for Model to Make Predictions on Test Set}}{\text{Number of Samples in Test Set}} \tag{6}$$

3.7 Computational Requirements

To facilitate the practical deployment of our CNN model and provide a clearer understanding of its operational demands, we have conducted an in-depth analysis of its computational requirements. The model's architecture, primarily designed for efficiency and scalability, has been evaluated in terms of memory usage, processing power, and execution time.

Our CNN model requires the following computational resources:

- **Memory Usage:** The model uses approximately 2 GB of GPU memory during training on datasets containing up to 200,000 tweets or comments. This memory footprint includes the storage of model weights, intermediate activations, and gradients during backpropagation.
- **Processing Power:** Training is optimized for CUDA-enabled GPUs. A minimum of an NVIDIA GTX 1060 GPU or equivalent is recommended for training within reasonable time frames. Training the model on a dataset of approximately 200,000 tweets takes about 2 hours on this class of GPU.
- **Runtime Metrics:** Inference on new data is relatively fast, with the model processing around 1,000 tweets per minute on a standard GPU setup. This performance metric ensures that the model can be deployed effectively in real-time applications where quick sentiment analysis is crucial.

4 Evaluation and Results

This section critically evaluates the performance of our proposed sentiment analysis model across different computational settings. By leveraging a dataset specifically curated to analyze sentiments expressed during a politically charged event, this study not only underscores the model's robustness but also its adaptability across varied batch sizes and epochs. The comprehensive analysis provides insights into the optimal configuration for deploying this model in real-world scenarios, balancing computational efficiency and predictive accuracy.

4.1 Dataset Description

The dataset utilized in this research [1] was developed specifically for analyzing sentiment across different social media platforms, focusing on public opinion regarding political figures during the 2019 general elections in India. Data was sourced from Twitter and Reddit using Tweepy and PRAW APIs, respectively, and consists of two distinct datasets: approximately 163K tweets and 37K comments from Reddit. Each piece of data was preprocessed using Python's re module to clean and standardize the text, removing URLs, hashtags, and mentions, while applying NLP techniques for tokenization and normalization. Sentiments were labeled on a scale from -1 (negative) to 1 (positive), with 0 indicating a neutral sentiment.

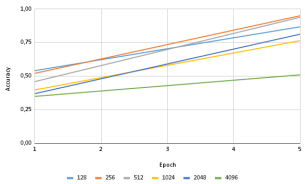
4.2 Experimental Setup and Results

Experiments were designed to evaluate the performance of our sentiment analysis model across different batch sizes and number of

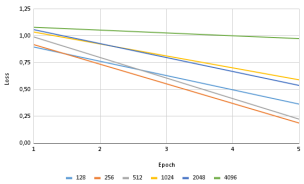
Table 2: Experimental Evaluation

Epochs	Loss	Accuracy	Time	Loss	Accuracy	Time	Loss	Accuracy	Time
	Batch Size = 128			**Batch Size = 256**			**Batch Size = 512**		
1	0.8954	0.5401	86	0.9179	0.5199	86	0.9892	0.4582	106
5	0.3637	0.8669	81	0.1872	0.9493	76	0.2235	0.9379	87
	Batch Size = 1024			**Batch Size = 2048**			**Batch Size = 4096**		
1	1.035	0.3965	98	1.057	0.3687	98	1.079	0.3487	107
5	0.5896	0.7639	92	0.5374	0.8119	93	0.9734	0.5090	92

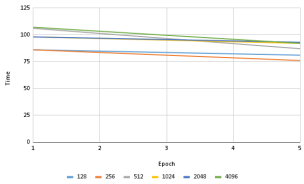
epochs. The key performance metrics assessed were accuracy, loss, and computational efficiency, with results summarized in Table 2 and visually depicted in Figure 1. We explored batch sizes ranging from 128 to 4096, focusing on their impact on model training dynamics.



(a) Accuracy



(b) Loss



(c) Time

Figure 1: Accuracy, Loss and Time for different Batch Sizes

The detailed examination of the numerical results reveals several key insights into the model's performance across varying batch sizes. Notably, the highest initial loss observed is 1.079 for the largest batch size of 4096, which suggests that larger batch sizes may struggle with initial model convergence. In contrast, the smallest batch size of 128 starts with a significantly lower initial loss of 0.8954, highlighting its more effective learning rate in the early stages of training. Additionally, the accuracy improvement from the first to the fifth epoch is most pronounced in the batch size of 256, where it surges from 0.5199 to 0.9493, marking the highest accuracy gain among all configurations. This specific batch size not only shows a substantial reduction in loss—from 0.9179 to 0.1872—but also manages to achieve this with a minimal decrease in training time, reducing only from 86 units to 76 units. These metrics underscore the efficiency of intermediate batch sizes in balancing computational demands with robust learning outcomes, suggesting their suitability for scenarios requiring both high accuracy and moderate resource utilization.

These findings suggest that the choice of batch size should be carefully considered based on the specific needs of the deployment environment and the desired balance between accuracy and training speed. Further research could investigate adaptive batch sizing or other optimization techniques to dynamically balance these factors during model training.

5 Conclusions and Future Work

This study has effectively demonstrated the capabilities of a Convolutional Neural Network (CNN) model tailored for sentiment analysis across multi-source social media platforms, specifically examining Twitter and Reddit data related to political discussions. Our experiments have systematically explored the impact of varying batch sizes on model performance, revealing that smaller batch sizes facilitate quicker learning and improved accuracy, albeit at the cost of increased computational demands. Notably, the batch size of 256 emerged as particularly effective, achieving the best balance between accuracy, loss reduction, and computational efficiency.

The findings from this research underscore the potential of CNNs in processing and analyzing textual data, reinforcing the model's utility in extracting meaningful insights from complex and voluminous datasets. The adaptability of the model across different computational settings further highlights its applicability in diverse operational environments, from resource-constrained systems to more robust computational frameworks.

Looking ahead, there are several promising avenues for extending this work. To address the gaps identified in our comparison analysis, future studies will include a comprehensive benchmarking of our CNN model against traditional sentiment analysis models such as Naive Bayes, Support Vector Machines, and Multi-Layer Perceptrons. This will allow us to provide a detailed evaluation of performance improvements and operational efficiencies gained through our deep learning approach.

Additionally, investigating the integration of adaptive batch sizing techniques could provide a way to dynamically optimize training processes based on real-time performance metrics, potentially enhancing both efficiency and accuracy. Experimenting with newer deep learning architectures such as Transformer models could offer improvements in handling long-range dependencies in text, which is particularly relevant for sentiment analysis involving nuanced political discourse.

Furthermore, we acknowledge the importance of model generalization in real-world applications. As such, future research will focus on evaluating the generalizability of our CNN model by applying it to varied datasets across different fields and languages. This expansion will involve systematic testing on datasets collected from domains such as customer service, health care, and entertainment, where sentiment analysis plays a crucial role. By doing

so, we aim to explore the model's adaptability and performance in environments that differ significantly from political discourse. Additionally, this approach will allow us to refine the model further, ensuring it can handle a broader spectrum of linguistic features and cultural contexts, thus enhancing its practical utility in diverse global settings.

Moreover, expanding the dataset to include multi-lingual and cross-cultural contexts could enhance the generalizability of the findings and provide deeper insights into the global applicability of the proposed model. This would be invaluable in developing more sophisticated NLP systems that are sensitive to cultural and linguistic nuances, which are crucial in global platforms like Twitter and Reddit.

In conclusion, while this study has made significant strides in advancing sentiment analysis via deep learning, the continuously evolving nature of online discourse presents ongoing challenges and opportunities for further research. Future work will need to focus on refining these models, enhancing their scalability and adaptability, and exploring their potential in wider applications beyond the scope of political sentiment analysis.

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